

Digital Material Passport

ID DMP-2024-ARR-001 - Version 0.1.0  
Issue Date 2024-12-15 - Certificate Type EN 10204 3.1

| Manufacturer                                                                                              | Customer                                                                                                             | Business Transaction |                       |
|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------|-----------------------|
| Advanced Steel Technologies GmbH<br>Industriestraße 15<br>52070 Aachen<br>DE<br>quality@advanced-steel.de | Precision Engineering Ltd.<br>Manufacturing Park 42<br>Birmingham<br>B1 1AA<br>GB<br>procurement@precision-eng.co.uk | Order                | PO-2024-7890          |
|                                                                                                           |                                                                                                                      | Delivery             | DN-2024-3456 - 500 kg |
|                                                                                                           |                                                                                                                      |                      |                       |
|                                                                                                           |                                                                                                                      |                      |                       |

Product

|              |                          |       |                                      |
|--------------|--------------------------|-------|--------------------------------------|
| Product Name | Hardenability Test Steel | Shape | RoundBar - D 100 mm - Length 1000 mm |
| Batch ID     | HTB-2024-045             |       |                                      |

Chemical Analysis

Heat Number H-2024-078 - Sample Location Ladle

| Symbol | C    | Mn   | Si   | P     | S     | Cr   | Ni   |
|--------|------|------|------|-------|-------|------|------|
| Unit   | %    | %    | %    | %     | %     | %    | %    |
| Min    | 0.42 | 0.50 |      |       |       |      |      |
| Max    | 0.50 | 0.80 | 0.40 | 0.030 | 0.025 | 0.25 | 0.25 |
| Actual | 0.45 | 0.65 | 0.25 | 0.015 | 0.008 | 0.12 | 0.08 |

Mechanical Properties

| Property                        | Symbol | Actual | Minimum | Maximum | Method  | Status |      |      |       |       |
|---------------------------------|--------|--------|---------|---------|---------|--------|------|------|-------|-------|
| Jominy Hardenability Test       |        |        |         |         | ISO 377 | ✓      |      |      |       |       |
| Quenched from 850°C             |        |        |         |         |         |        |      |      |       |       |
| Distance from quenched end (mm) | 1.5    | 3      | 5       | 7       | 9       | 11     | 15   | 20   | 25    | 30    |
| Min                             | 42     | 39     | 35      | 32      | 29      | 26     | 22   | 20   |       |       |
| Max                             | 47     | 46     | 44      | 41      | 39      | 37     | 33   | 31   | <= 30 | <= 29 |
| Value [HRC]                     | 45     | 43     | 40      | 36      | 33      | 31     | 27   | 25   | 28    | 27    |
| Status                          | Pass   | Pass   | Pass    | Pass    | Pass    | Pass   | Pass | Pass | Pass  | Pass  |

Coating Thickness Profile

|                            |        |  |        |          |        |   |
|----------------------------|--------|--|--------|----------|--------|---|
| Magnetic induction method  |        |  |        | ISO 2178 |        | ✓ |
| Position along length (mm) | 0      |  | 50     |          | 100    |   |
| Min                        | >= 120 |  | >= 110 |          | >= 100 |   |
| Max                        | <= 130 |  | <= 125 |          | <= 120 |   |
| Value [µm]                 | 125    |  | 115    |          | 108    |   |
| Status                     | Pass   |  | Pass   |          | Pass   |   |

Validation

We hereby certify that the material described above has been manufactured and tested in accordance with EN 10083-1 and ISO 377. All hardenability requirements have been met.

Validated by **Dr. Maria Schmidt**, Metallurgical Engineer,  
Quality Control Laboratory - 2024-12-15